

*MA.7.DP.2.1***Benchmark****MA.7.DP.2.1 Determine the sample space for a simple experiment.**Benchmark Clarifications:

Clarification 1: Simple experiments include tossing a fair coin, rolling a fair die, picking a card randomly from a deck, picking marbles randomly from a bag and spinning a fair spinner.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.2.2
- MA.7.DP.2.3, MA.7.DP.2.4

Terms from the K-12 Glossary

- Event
- Sample Space

Vertical Alignment**Previous Benchmarks**

- This is the first introduction to sample space.

Next Benchmarks

- MA.8.DP.2.1, MA.8.DP.2.2, MA.8.DP.2.3

Purpose and Instructional Strategies

In grade 7, students determine the sample space for a simple experiment. In grade 8, they will find the sample space for a repeated experiment.

- For mastery of this benchmark, an experiment is an action that can have more than one outcome. Experiments tend to have randomness, or uncertainty, in their outcomes.
 - For example, an experiment can be the action of tossing a coin. Possible outcomes would be whether the coin lands on heads or lands on tails.
- For mastery of this benchmark, simple experiments are restricted to those listed in *Clarification 1*.
 - Tossing a coin
Coins are not limited to those with heads or tails.
 - Rolling a die
Dice are not limited to 6-sided dice.
 - Picking a card from a deck
Card decks are not limited to a standard 52-card deck.
 - Picking a marble from a bag
Picking a marble from a bag is not limited to colors. Picking a tile, slip of paper or other objects from a bag are acceptable for this benchmark.
 - Spinning a spinner
Spinning a spinner is not limited to colors.
- Students should experience experiments before discussing the theoretical concept of probability.

- Students should informally explore the idea of likelihood, fairness and chance while building the meaning of a probability value. In this benchmark, all experiments are fair, meaning that all of the individual outcomes are equally likely.
 - For example, if the experiment is to draw a marble from a bag, then each marble is equally likely to be chosen.
- Have students practice making models to represent sample spaces to gain understanding on how probabilities are determined. Use familiar tools, including virtual manipulatives such as a coin, fair die, deck of cards and fair spinner (*MTR.2.1*).
- For simple experiments, a sample space will typically be represented by a list of outcomes, such as Heads, Tails or by a written description, such as “The sides of a 20-sided die.” Providing opportunities for students to match situations and sample spaces will assist with building their ability to visualize the sample space for any given experiment.
 - For example, the experiment of drawing a marble from a bag containing 2 red marbles and 1 blue marble has the sample space that can be written as {red, red, blue} or as {r, r, b}.

Common Misconceptions or Errors

- Students may incorrectly list more outcomes than the experiment merits. To address this misconception, ensure students can explain the experiment in their own words to then verify what is listed in their sample space.

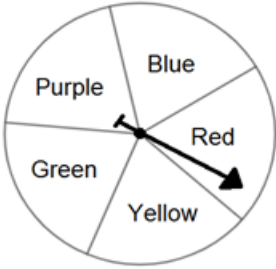
Strategies to Support Tiered Instruction

- Teacher provides examples of situations and has students decide on the sample space necessary.
- Teacher co-creates a graphic organizer with different examples of sample spaces with the use of virtual manipulatives.
- Teacher co-creates a graphic organizer of a T-chart to list the experiment and the sample space necessary for the examples provided.
- Teacher co-creates models with students to represent sample spaces using a coin, fair die, fair spinner or deck of cards.
- Teacher ensures students can explain the experiment in their own words to then verify what is listed in their sample space.

Instructional Tasks

Instructional Task 1 (MTR.4.1)

List all of the possible outcomes for each experiment.

Experiment	Sample Space
rolling a 12-sided fair die	
flipping a coin	
pulling a face card from a standard deck of cards	
a spin from the spinner 	

Compare your list with a partner and identify any differences. Allow each partner time to discuss their reasoning until an agreement is reached on the correct sample space.

Instructional Items

Instructional Item 1

Letter cards for the word “probability” are placed into a bag. List the sample space for choosing a card from this bag.

Instructional Item 2

There are 10 blue, 5 green and 7 white marbles in a jar. List the sample space for the simple experiment of choosing a marble from the jar.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*